RAJALAKSHMI ENGINEERING COLLEGE

Ameera Fathima M 250701048-2025-CSE A2

Started on	Thursday, 13 August 2026, 3:58 PM
State	Finished
Completed on	Thursday, 13 August 2026, 4:01 PM
Time taken	3 mins 15 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include <stdio.h>
void function(int n) {
    int count = 0;
    int c = 0;
    count++; // 1 step for the initialization statement:
    count++; // For the initial/final False evaluation of the condition (i < n)
    for (int i = n / 2; i < n; i++) {
        count++; // For each successful True evaluation of (i < n)
        count++; // For the initial/final False evaluation of the condition (j < n)
        for (int j = 1; j < n; j = 2 * j) {
            count++; // For each successful True evaluation of (j < n)
            count++; // For the initial/final False evaluation of the condition (k < n)
            for (int k = 1; k < n; k = k * 2) {
                count++; // For each successful True evaluation of (k < n)
                count++; // 1 step for the execution of the loop body statement (c++)
                c++;                }        }    }
                printf("%d", count); }
    int main() {
        int n;
        scanf("%d", &n);
        function(n);
        return 0; }
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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